

THE LIFTING PROBLEM FOR UNIVERSAL QUADRATIC FORMS IN DEGREES SIX AND SEVEN

ABSTRACT. We show that no totally real number field of degree 6 or 7 admits a universal quadratic form with rational integer coefficients (a universal $\mathbb{Z}$ -form). Together with the classification in degrees ≤ 5 by Kala and Yatsyna, the fields of degree at most 7 with a universal $\mathbb{Z}$ -form are exactly $\mathbb{Q}$, $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$. The proof replaces the house bound of the degree- ≤ 5 argument by the geometry of the *variance lattice* $\Lambda_K = \mathcal{O}_K/\mathbb{Z}$: a tensor-product minimum bound turns the existence of a universal $\mathbb{Z}$ -form into lower bounds for traces of totally positive integers, and a “downspread” argument bounds the shortest vector of Λ_K by $(5 + \sqrt{21})/2$. This makes the search space small: in the case where the shortest vector generates K , no number field has to be computed at all (every candidate dies inside the order $\mathbb{Z}[x_1]$); in the imprimitive case only 18 fields are ever built, and the last five are excluded by explicit elements. A second, independent route through the Mordell–Ko theorem and the E_6 trace threshold confirms every final certificate. In degree 7 (no proper subfields, $\nu(K) \leq (11 + \sqrt{105})/4$) all $5.3 \cdot 10^8$ candidate polynomials are excluded inside $\mathbb{Z}[x_1]$, again without computing a single number field.

1. STATEMENT AND OVERVIEW

Let K be a totally real number field of degree d with ring of integers $\mathcal{O}_K$. A $\mathbb{Z}$ -form is a positive definite quadratic form $Q(x) = \sum_{i \leq j} a_{ij} x_i x_j$ with $a_{ij} \in \mathbb{Z}$; it is *universal over K* if it represents every totally positive element of $\mathcal{O}_K$ over $\mathcal{O}_K$. Kala and Yatsyna [KY2] proved that for $d \leq 5$ such a form exists only for $K = \mathbb{Q}, \mathbb{Q}(\sqrt{5}), \mathbb{Q}(\zeta_7 + \zeta_7^{-1})$. Their proof uses the Mordell–Ko theorem [Ko]: in rank ≤ 5 every positive definite classical integral form is a sum of squares of integral linear forms. This fails in rank 6 (the lattice E_6), which is why degree 6 was the first open case.

Theorem 1.1. *No totally real number field of degree 6 admits a universal $\mathbb{Z}$ -form.*

Theorem 1.2. *No totally real number field of degree 7 admits a universal $\mathbb{Z}$ -form.*

Corollary 1.3. *A totally real field of degree $d \leq 7$ admits a universal $\mathbb{Z}$ -form if and only if it is $\mathbb{Q}$, $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$.*

The proof has a theoretical part (§§2–5), which reduces the theorem to a finite computation with explicit bounds, and a computational part (§6 for degree 6, §7 for degree 7). The main steps are:

- (i) The necessary condition (N) of [KY1]: every totally positive α is $\alpha = \sum M_{ij} \omega_i \omega_j$ for a positive semidefinite half-integral M .
- (ii) A tensor-product bound (Lemma 3.2) turns (N) into a *budget–cost* inequality $\tau(\alpha) \geq \nu(K)$ (Proposition 3.3) and into a *rigidity* statement for elements with $\tau(\alpha) < \frac{3}{2}\nu(K)$ (Proposition 3.4), plus relative versions over subfields.
- (iii) Applying budget–cost to $x_1 - n$ and $-x_1 - n'$, for x_1 a shortest vector of the variance lattice, gives $\nu(K) \leq (5 + \sqrt{21})/2$ (Corollary 4.2). Over a proper subfield, the same argument together with a “positive covering constant” δ_F bounds the relative minimum (Corollary 4.6).
- (iv) The resulting finite enumerations are run in §6. Every candidate field is excluded, and every exclusion is witnessed by an explicit element.

2. SETUP

Throughout, K is totally real of degree d with embeddings $\sigma_1, \dots, \sigma_d$. For $x \in K$ put

$$\tau(x) = \frac{1}{d} \operatorname{Tr}_{K/\mathbb{Q}}(x), \quad \operatorname{Var}(x) = \frac{1}{d} \sum_{j=1}^d (\sigma_j(x) - \tau(x))^2.$$

Both are intrinsic: if x lies in a subfield F , they coincide with the quantities computed in F . The *variance lattice* is $\Lambda_K = \mathcal{O}_K/\mathbb{Z}$ with the positive definite quadratic form $d \operatorname{Var}$; it is a lattice of rank $d-1$ (the embedding $x \mapsto (\sigma_j(x) - \tau(x))_j$ identifies it with a lattice in $\mathbf{1}^\perp \subset \mathbb{R}^d$). Put

$$\nu(K) = \min\{\operatorname{Var}(x) : x \in \mathcal{O}_K \setminus \mathbb{Z}\}, \quad t(K) = \inf\{\tau(\alpha) : \alpha \in \mathcal{O}_K^+ \setminus \mathbb{Z}\}.$$

A *shortest vector* is an $x_1 \in \mathcal{O}_K \setminus \mathbb{Z}$ with $\operatorname{Var}(x_1) = \nu(K)$; it is determined by its class up to $x_1 \mapsto \pm x_1 + k$ ($k \in \mathbb{Z}$).

For a subfield $F \subseteq K$ with $f = [F : \mathbb{Q}]$ and $e = [K : F]$, group the embeddings of K into f blocks $B_1, \dots, B_f$ of size e (the fibres over the embeddings $\rho_1, \dots, \rho_f$ of F), and define the block means $\mu_k(x) = \frac{1}{e} \sum_{j \in B_k} \sigma_j(x)$ and

$$\operatorname{Var}_{K/F}(x) = \frac{1}{d} \sum_{k=1}^f \sum_{j \in B_k} (\sigma_j(x) - \mu_k(x))^2, \quad \nu_{K/F} = \min\{\operatorname{Var}_{K/F}(x) : x \in \mathcal{O}_K \setminus F\}.$$

The map $x \mapsto (\sigma_j(x) - \mu_k(x))_{j \in B_k, k}$ has kernel F , and $\mathcal{O}_K/\mathcal{O}_F$ is torsion free, so $\operatorname{Var}_{K/F}$ is a positive definite form on the lattice $\mathcal{O}_K/\mathcal{O}_F$ of rank $d-f$.

3. CONDITION (N) AND THE TENSOR BOUND

Lemma 3.1 ([KY1, Prop. 3.1]). *Let $\omega_1, \dots, \omega_d$ be a $\mathbb{Z}$ -basis of $\mathcal{O}_K$. If K admits a universal $\mathbb{Z}$ -form, then every $\alpha \in \mathcal{O}_K^+$ satisfies*

$$(N) \quad \alpha = \sum_{i,j} M_{ij} \omega_i \omega_j \quad \text{for some positive semidefinite } M = M^t \text{ with } M_{ii} \in \mathbb{Z}, \ 2M_{ij} \in \mathbb{Z}.$$

Proof. If Q has Gram matrix M_Q and $\alpha = Q(v)$ with $v \in \mathcal{O}_K^r$, write $v = V\omega$ with $V \in \mathbb{Z}^{r \times d}$ and take $M = V^t M_Q V$. $\square$

Equivalently, $2\alpha = Q_L(v)$ where $L = (\mathbb{Z}^d, 2M)$ is an *even* lattice. Replacing L by its quotient by the radical, we may take L positive definite of rank $\leq d$; then $\min L \geq 2$, and every vector of L that is not a minimal vector has norm ≥ 4 .

Lemma 3.2 (tensor floor). *Let L, M be positive definite lattices (real quadratic forms on free $\mathbb{Z}$ -modules), and let $w \in L \otimes M$, $w \neq 0$, have tensor rank s . Then $Q(w) \geq \frac{s}{\gamma_s^2} \min L \cdot \min M$, where γ_s is Hermite's constant. In particular, for $s \leq 6$,*

$$Q(w) \geq \min L \cdot \min M, \quad \text{and} \quad Q(w) \geq \frac{3}{2} \min L \cdot \min M \quad \text{if } s \geq 2.$$

Proof. By the Smith normal form, the coefficient matrix of w is $\sum_{i=1}^s x_i y_i^t$ with $x_1, \dots, x_s \in L$ and $y_1, \dots, y_s \in M$ linearly independent. With the Gram matrices X of the x_i and Y of the y_i we have $Q(w) = \operatorname{tr}(XY) = \operatorname{tr}(X^{1/2} Y X^{1/2}) \geq s(\det X \det Y)^{1/s}$ by the arithmetic-geometric mean inequality, and $\det X \geq (\min L / \gamma_s)^s$, $\det Y \geq (\min M / \gamma_s)^s$ by Hermite's inequality for the lattices spanned by the x_i and the y_i . The known values $\gamma_1^2 = 1$, $\gamma_2^2 = \frac{4}{3}$, $\gamma_3^2 = 2^{2/3}$, $\gamma_4^2 = 2$, $\gamma_5^2 = 8^{2/5}$, $\gamma_6^2 = (64/3)^{1/3}$ (see [CS]) give $s/\gamma_s^2 = 1, 1.5, 1.889, 2, 2.176, 2.163$ for $s = 1, \dots, 6$. $\square$

This is the elementary case of Kitaoka's theorem on minima of tensor products [Ki]; we only need tensor rank ≤ 6 , because $\operatorname{rk} \Lambda_K = d-1 \leq 5$ when $d \leq 6$.

Proposition 3.3 (budget-cost). *Let $d \leq 7$ and suppose K admits a universal $\mathbb{Z}$ -form. Then $\tau(\alpha) \geq \nu(K)$ for every $\alpha \in \mathcal{O}_K^+ \setminus \mathbb{Z}$; that is, $t(K) \geq \nu(K)$.*

Proof. Write $2\alpha = Q_L(v)$ with $v \in L \otimes \mathcal{O}_K$ as above, and let $m = \frac{1}{d} \sum_j \sigma_j(v) \in L \otimes \mathbb{R}$. Averaging $Q_L(\sigma_j v) = Q_L(m) + 2B(m, \sigma_j v - m) + Q_L(\sigma_j v - m)$ over j ,

$$(1) \quad 2\tau(\alpha) = Q_L(m) + \frac{1}{d} Q_{L \otimes \Lambda_K}(\bar{v}),$$

where $\bar{v}$ is the image of v in $L \otimes \Lambda_K$ and Λ_K carries the form $d \text{Var}$. Since $\alpha \notin \mathbb{Z}$ we have $v \notin L \otimes \mathbb{Z}$, so $\bar{v} \neq 0$ ($\mathbb{Z}$ is saturated in $\mathcal{O}_K$), and Lemma 3.2 gives $Q_{L \otimes \Lambda_K}(\bar{v}) \geq 2 \cdot d\nu(K)$. $\square$

Proposition 3.4 (rigidity). *Under the hypotheses of Proposition 3.3, let $\alpha \in \mathcal{O}_K^+ \setminus \mathbb{Z}$ with $\tau(\alpha) < \frac{3}{2}\nu(K)$. Then $\alpha = y^2 + cy + b$ with $y \in \mathcal{O}_K \setminus \mathbb{Z}$, $b, c \in \mathbb{Z}$ and $\nu(K) \leq \text{Var}(y) \leq \tau(\alpha)$.*

Proof. In (1), $\frac{1}{d}Q(\bar{v}) < 3\nu(K)$. By Lemma 3.2, $\bar{v}$ has tensor rank 1, i.e. $\bar{v} = e \otimes \bar{y}$, and $Q_L(e) = 2$ (otherwise $Q_L(e) \geq 4$ and the cost is $\geq 4\nu$). Lifting, $v = e \otimes y + r$ with $r \in L$, so $2\alpha = 2y^2 + 2B(e, r)y + Q_L(r)$ with $Q_L(r) \in 2\mathbb{Z}$. Finally $2\tau(\alpha) \geq \frac{1}{d}Q(\bar{v}) = 2 \text{Var}(y)$. $\square$

Proposition 3.5 (relative versions). *Suppose K ($d \leq 6$) admits a universal $\mathbb{Z}$ -form, and let $F \subsetneq K$ be a subfield. Then $\tau(\alpha) \geq \nu_{K/F}$ for every $\alpha \in \mathcal{O}_K^+ \setminus F$. If moreover $\tau(\alpha) < \frac{3}{2}\nu_{K/F}$, then $\alpha = y^2 + \gamma y + \beta$ with $y \in \mathcal{O}_K \setminus F$, $\beta, \gamma \in \mathcal{O}_F$ and $\text{Var}_{K/F}(y) \leq \tau(\alpha)$.*

Proof. Decompose $\mathbb{R}^d$ orthogonally into block-constant vectors and their complement. The part of $\frac{1}{d}Q(\bar{v})$ in the complement is $\frac{1}{d}Q(\pi\bar{v})$, where $\pi\bar{v} \in L \otimes \pi(\Lambda_K)$ and $\pi(\Lambda_K) \cong \mathcal{O}_K/\mathcal{O}_F$ carries the form $d \text{Var}_{K/F}$ of minimum $d\nu_{K/F}$. If $\alpha \notin F$ then $v \notin L \otimes \mathcal{O}_F$, so $\pi\bar{v} \neq 0$; apply Lemma 3.2 as before. For rigidity, $\pi\bar{v} = e \otimes \pi(\bar{y})$ with $Q_L(e) = 2$. Because $\ker \pi$ on $L \otimes \Lambda_K$ is $L \otimes \Lambda_F$, this gives $v - e \otimes y \in L \otimes \mathcal{O}_F$; expand as before, using that L is even. $\square$

4. BOUNDING THE MINIMA

Lemma 4.1 (downspread). *Let $x \in \mathcal{O}_K \setminus \mathbb{Z}$ with conjugates $\theta_1 < \dots < \theta_d$. Then $\alpha_- = x - (\lceil \theta_1 \rceil - 1)$ and $\alpha_+ = -x + (\lfloor \theta_d \rfloor + 1)$ are totally positive and non-rational, and $\tau(\alpha_-) = \tau(x) - \lceil \theta_1 \rceil + 1 < \tau(x) - \theta_1 + 1$, $\tau(\alpha_+) = \lfloor \theta_d \rfloor + 1 - \tau(x) < \theta_d - \tau(x) + 1$.*

Corollary 4.2. *If K of degree $d \leq 7$ admits a universal $\mathbb{Z}$ -form, then $d\nu \geq 2(\nu - 1)^2$ for $\nu = \nu(K)$. In particular $\nu(K) \leq \frac{(d+4) + \sqrt{(d+4)^2 - 16}}{4}$, which for $d = 6$ is $(5 + \sqrt{21})/2 = 4.7913\dots$*

Proof. Let x_1 be a shortest vector and $w_j = \sigma_j(x_1) - \tau(x_1)$. By Proposition 3.3 and Lemma 4.1, $-\min_j w_j > \nu - 1$ and $\max_j w_j > \nu - 1$. Hence $d\nu = \sum w_j^2 \geq 2(\nu - 1)^2$. $\square$

The enumeration of §6 uses the exact forms of these inequalities (with the integer parts), together with the following consequence of budget-cost for quadratic polynomials in x_1 .

Lemma 4.3. *Let x_1 be a shortest vector of degree ≥ 3 and $P \in \mathbb{Z}[t]$ monic of degree 2 with $P(\sigma_j(x_1)) > 0$ for all j . Under the hypotheses of Proposition 3.3, $P(\tau(x_1)) \geq 0$.*

Proof. $P(x_1)$ is totally positive and not rational, and $\tau(P(x_1)) = \text{Var}(x_1) + P(\tau(x_1))$. $\square$

Definition 4.4. For a totally real field F of degree f with Minkowski lattice $\sigma(\mathcal{O}_F) \subset \mathbb{R}^f$, the positive covering constant is

$$\delta_F = \sup_{u \in \mathbb{R}^f} \min \left\{ \frac{1}{f} \sum_k (p_k - u_k) : p \in \sigma(\mathcal{O}_F), p_k > u_k \text{ for all } k \right\}.$$

For example $\delta_{\mathbb{Q}} = 1$. The function $h(u) = \min_{p > u} \sum_k (p_k - u_k)$ is $\sigma(\mathcal{O}_F)$ -periodic and satisfies $h(u) \leq h(u') + \sum_k (u'_k - u_k)$ whenever $u \leq u'$ coordinatewise. Evaluating h on a grid of step ε that covers a fundamental parallelepiped therefore gives the rigorous bound $\delta_F \leq \max_{\text{grid}} h/f + \varepsilon$ (deltaF.py).

Lemma 4.5 (relative downspread). *For $y \in \mathcal{O}_K \setminus F$ with block minima $m_k = \min_{j \in B_k} \sigma_j(y)$, there is $\gamma \in \mathcal{O}_F$ with $y + \gamma$ totally positive and $\tau(y + \gamma) \leq \frac{1}{f} \sum_k (\mu_k(y) - m_k) + \delta_F$. The same holds for $-y$ with the block maxima.*

Proof. $\tau(y + \gamma) = \frac{1}{f} \sum_k (\mu_k(y) - m_k) + \frac{1}{f} \sum_k (\rho_k(\gamma) + m_k)$, and the second sum can be made $\leq f\delta_F$ with all $\rho_k(\gamma) + m_k > 0$. $\square$

Corollary 4.6. *If K ($d \leq 6$) admits a universal $\mathbb{Z}$ -form and $F \subsetneq K$ with $e = [K : F]$, then $e\nu_{K/F} \geq 2(\nu_{K/F} - \delta_F)^2$.*

Proof. Let y realise $\nu_{K/F}$, and write u_k for the vector of within-block deviations in block k , with maximum M_k and minimum $-m'_k$. By Proposition 3.5 and Lemma 4.5, $D_{\pm} \geq \nu_{K/F} - \delta_F$, where $D_+ = \frac{1}{f} \sum M_k$ and $D_- = \frac{1}{f} \sum m'_k$. Since $\|u_k\|^2 \geq M_k^2 + m_k'^2 \geq (M_k + m'_k)^2/2$, $d\nu_{K/F} = \sum_k \|u_k\|^2 \geq \frac{1}{2f} (\sum_k (M_k + m'_k))^2 \geq 2f(\nu_{K/F} - \delta_F)^2$. $\square$

For $F = \mathbb{Q}$ this is Corollary 4.2 (with $\delta_{\mathbb{Q}} = 1$, $e = d$).

5. RIGIDITY VIA SQUARE ROOTS, AND THE E_6 THRESHOLD

For testing, rigidity is used in the following form, which needs no search in $\mathcal{O}_K$.

Lemma 5.1. *Let α be as in Proposition 3.4 and suppose $\mathbb{Q}(\alpha) = K$. Then there is $\kappa \in \mathbb{Z}$ with $\kappa \equiv 0, 3 \pmod{4}$ such that $4\alpha - \kappa = z^2$ with $z \in \mathcal{O}_K$, $4\nu(K) \leq \text{Var}(z) \leq 4\tau(\alpha)$ and $0 < 4 \min_j \sigma_j(\alpha) - \kappa \leq \max(\frac{d^2}{d-1} \tau(\alpha), \text{Var}(\alpha)/\nu(K))$ (for $d = 6$: $\frac{36}{5} \tau(\alpha) = 7.2 \tau(\alpha)$). Equivalently, the polynomial $4^d m_{\alpha}((t^2 + \kappa)/4) \in \mathbb{Z}[t]$ has an irreducible factor of degree d whose roots have variance in $[4\nu(K), 4\tau(\alpha)]$.*

Proof. Take $z = 2y + c$ and $\kappa = 4b - c^2$. Put $u = 4 \min \sigma(\alpha) - \kappa$, so that $|\sigma_j(z)| \geq \sqrt{u}$. If $p \geq 1$ conjugates of z are positive and $q \geq 1$ negative, then $\text{Var}(z) \geq \frac{pq}{d^2} (2\sqrt{u})^2 \geq \frac{d-1}{d^2} 4u$, so $u \leq \frac{d^2}{d-1} \tau(\alpha)$. If they all agree, $\text{Var}(z) = \text{Var}(\sqrt{4\alpha - \kappa}) \leq 4 \text{Var}(\alpha)/u$, and $\text{Var}(z) \geq 4\nu$ gives $u \leq \text{Var}(\alpha)/\nu$. $\square$

The second tool is used as an independent cross-check of the final certificates. It is the place where E_6 enters.

Proposition 5.2 (E_6 threshold). *Let $d = 6$ and $\alpha \in \mathcal{O}_K^+$ with $\text{Tr } \alpha < 3(3d_K)^{1/6}$. Then (N) holds for α if and only if*

$$(\text{SoS}_2) \quad 2\alpha = \sum_k x_k^2 \quad \text{with } x_k \in \mathcal{O}_K \quad \text{and} \quad \sum_k x_k \in 2\mathcal{O}_K.$$

Proof. If (SoS₂) holds, write $x = U\omega$ and take $M = \frac{1}{2}U^t U$: its diagonal is integral because $\sum_k U_{ki}^2 \equiv \sum_k U_{ki} \pmod{2}$. Conversely, let $2\alpha = Q_L(v)$ with $\tilde{L}$ even of rank $r \leq 6$, and write $v = \sum_k f_k \otimes \omega_k$ with $f_k \in L$. If $f_1, \dots, f_6$ are linearly dependent, then 2α is represented by an even lattice of rank ≤ 5 ; by Mordell–Ko [Ko, Mo] its Gram matrix is $U^t U$ with U integral, which gives (SoS₂). Otherwise $G = (B(f_k, f_l))$ is the Gram matrix of an even positive definite lattice of rank 6. Such a lattice has determinant ≥ 3 : even unimodular positive lattices have rank $\equiv 0 \pmod{8}$ [Se, Ch. V], and an even lattice of rank 6 and determinant 2 has discriminant form $q \equiv \frac{1}{2}$ or $\frac{3}{2} \pmod{2}$; gluing it with E_7 (discriminant form $\frac{3}{2}$), respectively with A_1 (discriminant form $\frac{1}{2}$), would produce an even unimodular lattice of rank 13, respectively 7. Hence $2 \text{Tr } \alpha = \text{tr}(GT) \geq 6(\det G \det T)^{1/6} \geq 6(3d_K)^{1/6}$, where T is the trace form of the basis ($\det T = d_K$). $\square$

The extremal case of the determinant bound is E_6 (determinant 3). It is the only obstruction to the degree- ≤ 5 method in degree 6, but it lives above trace $3(3d_K)^{1/6} \geq 29.47$. Every certificate below has trace ≤ 14 .

6. THE COMPUTATION

All code is in `higher-degree/src.run_degree6.sh` reproduces everything in about 10 minutes on 4 cores; `run_sanity.sh` runs the same machinery in degrees 2–5 (gcc, PARI/GP 2.15, python3 with numpy). Floating point is used only to *locate* candidates, always with margins in the direction that keeps more candidates. Every exclusion is decided in exact arithmetic.

Let K be a sextic field with a universal $\mathbb{Z}$ -form and x_1 a shortest vector. Either $\mathbb{Q}(x_1) = K$ (Case A) or $F = \mathbb{Q}(x_1)$ is a quadratic or cubic subfield (Case B).

6.1. Case A: the shortest vector generates K . Normalise $x_1 \mapsto \pm x_1 + k$ so that $\text{Tr } x_1 \in \{0, 1, 2, 3\}$. By Corollary 4.2, the minimal polynomial $x^6 + a_1x^5 + \cdots + a_6$ of x_1 satisfies $a_1 \in \{0, -1, -2, -3\}$ and $\sum_j (\sigma_j x_1 - \tau x_1)^2 \leq 15 + 3\sqrt{21}$. `enum6.c` enumerates all real-rooted integral polynomials with these constraints. It uses Robinson’s method: by Rolle’s theorem each coefficient is confined to an interval determined by the roots of the previous derivative [Ro, MS]. The program keeps only those whose roots pass the exact downspread inequalities of Lemma 4.1 (with $\nu = \text{Var}(x_1)$) and Lemma 4.3.

real-rooted polynomials enumerated (leaves)	2 088 667
passing downspread and quadratic filters	40 764
irreducible	15 720
NS: a shorter non-rational element exists in $\mathbb{Z}[x_1]$	873
BC: some $\alpha \in \mathbb{Z}[x_1]^+ \setminus \mathbb{Z}$ has $\tau(\alpha) < \text{Var}(x_1)$	5 062
RG: some $\alpha \in \mathbb{Z}[x_1]^+$ in the rigidity window fails Lemma 5.1	9 785
survivors	0

Each of the three order-level tests is a necessary condition for x_1 to be a shortest vector of Λ_K for a field with a universal $\mathbb{Z}$ -form. They are stated in the order $\mathbb{Z}[x_1] \subseteq \mathcal{O}_K$, so no maximal order is ever computed in Case A. The list of the 15 720 polynomials with the reason for each exclusion is `results/prim_status.txt`.

6.2. Case B: the shortest vector lies in a proper subfield. Now $\nu(K) = \nu(F)$, and every element of F is an element of K with the same τ and Var . The same enumeration in degrees 2 and 3 (in degree 2 without Lemma 4.3, which needs $\deg x_1 \geq 3$), using only the NS and BC tests (not rigidity, since the y of Proposition 3.4 may lie outside F), leaves exactly

$$F \in \{\mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\zeta_7)^+, \mathbb{Q}(\zeta_9)^+\}, \quad \nu(F) = \frac{5}{4}, 2, \frac{14}{9}, 2.$$

For each F let y be a relative shortest vector, so $K = F(y)$ since $[K : F] \in \{2, 3\}$ is prime. Corollary 4.6 and the certified bounds for δ_F give:

F	e	$\delta_F \leq$	$\nu_{K/F} \leq$	candidates	order-level survivors
$\mathbb{Q}(\sqrt{5})$	3	1.3110	3.6513	2504 relative cubics	20
$\mathbb{Q}(\sqrt{2})$	3	1.7091	4.2272	3873 relative cubics	34
$\mathbb{Q}(\zeta_7)^+$	2	1.5513	3.3934	1582 values of Δ	1
$\mathbb{Q}(\zeta_9)^+$	2	2.1529	4.2030	2242 values of Δ	7

Here a relative cubic is $y^3 + b_1y^2 + b_2y + b_3$ over $\mathcal{O}_F$, with b_1 modulo $3\mathcal{O}_F$ and $\frac{1}{6} \text{Tr}_F(\frac{2}{3}b_1^2 - 2b_2) = \text{Var}_{K/F}(y)$. A relative quadratic is $y = (-b_1 + \sqrt{\Delta})/2$ with $\Delta \in \mathcal{O}_F^+$, b_1 modulo $2\mathcal{O}_F$, $\Delta \equiv b_1^2 \pmod{4}$ and $\text{Var}_{K/F}(y) = \text{Tr}_F(\Delta)/12$. The order-level tests in $\mathcal{O}_F[y]$ are the absolute ones (with $\nu(K) = \nu(F)$) together with the relative budget–cost and “ y is relatively shortest”. The 62 survivors give 18 distinct fields. Only these 18 fields are built in the whole proof. On $\mathcal{O}_K$, 12 of them fail relative rigidity (Proposition 3.5) and one fails absolute rigidity. The remaining five fail (N) at an explicit element:

K	d_K	$\text{Tr } \alpha$	$N\alpha$	minimal polynomial of the witness α
$\mathbb{Q}(\sqrt{5}, \zeta_7^+)$	300125	14	29	$x^6 - 14x^5 + 73x^4 - 182x^3 + 227x^2 - 133x + 29$
$\mathbb{Q}(\sqrt{2}, \zeta_7^+)$	1229312	12	8	$x^2 - 4x + 2$ ($\alpha = 2 - \sqrt{2}$)
$\mathbb{Q}(\sqrt{2}, \zeta_9^+)$	3359232	12	8	$x^2 - 4x + 2$ ($\alpha = 2 - \sqrt{2}$)
$x^6 - 9x^4 + 24x^2 - 17$	7138368	12	9	$x^3 - 6x^2 + 9x - 3$ ($\alpha = 2 - \zeta_9 - \zeta_9^{-1}$)
$x^6 - 9x^4 + 24x^2 - 19$	7978176	12	9	$x^3 - 6x^2 + 9x - 3$ ($\alpha = 2 - \zeta_9 - \zeta_9^{-1}$)

Each failure of (N) is established by the exhaustive search of `condN.gp`. It uses $\text{tr } M \leq \text{Tr } \alpha / \lambda_{\min}(T)$, the bound $M_{ii} \leq (\sum_j |\sigma_j(\omega_i^*)| \sqrt{\sigma_j \alpha})^2$ with ω^* the trace-dual basis, integral solution of the linear system for the off-diagonal entries, a Fincke–Pohst enumeration of the solution coset, and an exact positive-semidefiniteness test. The witnesses are checked a second time by Proposition 5.2, with a different search (`sos2.gp`). In the last two fields there is no $x \neq 0$ with $x^2 \preceq 2\alpha$ at all, and in $\mathbb{Q}(\sqrt{5}, \zeta_7^+)$ the only such x are ± 1 . The first witness coincides with the one found earlier by independent code in the working pack [Pack].

6.3. Proof of Theorem 1.1. Suppose K is sextic with a universal $\mathbb{Z}$ -form and let x_1 be a shortest vector. In Case A, the minimal polynomial of a suitable normalisation of x_1 is among the 15 720 polynomials of §6.1, and each of them violates a condition that a shortest vector must satisfy. In Case B, $F = \mathbb{Q}(x_1)$ is one of the four fields listed, and $K = F(y)$ for a relative shortest vector y within the enumerated range. The resulting field is one of the 18 fields of §6.2, each of which violates rigidity or (N). $\square$

7. DEGREE SEVEN

Let K be a septic field with a universal $\mathbb{Z}$ -form. Budget–cost, rigidity and Corollary 4.2 hold for $d = 7$: the tensor rank is at most $\text{rk } \Lambda_K = 6$, and $s/\gamma_s^2 \geq \frac{3}{2}$ for $2 \leq s \leq 6$. Thus $\nu(K) \leq (11 + \sqrt{105})/4 = 5.3117\dots$ Since 7 is prime, K has no proper subfields. So the shortest vector x_1 generates K (only Case A occurs), and every non-rational element of K generates K . In particular Lemma 5.1 applies to every totally positive non-rational α in the rigidity window.

The enumeration of §6.1 (with $\text{Tr } x_1 \in \{0, 1, 2, 3\}$ and centred $T_2 \leq 7(11 + \sqrt{105})/4$) produces $5.3 \cdot 10^8$ leaves. So we also run two of the order-level tests inside the enumerator (`enum6.c`, mode `strong`):

- (NS) for $y = cx_1^2 + bx_1$ ($1 \leq c \leq 3$) and $y = x_1^3 + ax_1^2 + bx_1$, with the integers a, b near the real minimiser of $\text{Var}(y)$: if $0 < \text{Var}(y) < \text{Var}(x_1)$, then x_1 is not a shortest vector;
- (RIG) Lemma 5.1 for the two downspread elements $\alpha_{\pm}$ of Lemma 4.1, whenever $\tau(\alpha_{\pm}) < \frac{3}{2} \text{Var}(x_1)$. As $K = \mathbb{Q}(\alpha)$, the conjugates of z are $\pm \sqrt{4\sigma_j(\alpha) - \kappa}$. So $z \in \mathcal{O}_K$ exists iff, for some κ in the range of the lemma and some choice of signs, all elementary symmetric functions of these numbers are integers, and $\text{Var}(z) \in [4 \text{Var}(x_1), 4\tau(\alpha)]$.

real-rooted polynomials enumerated (leaves)	533 562 406
excluded by the downspread inequalities (Lemma 4.1)	531 897 850
excluded by Lemma 4.3	41 498
excluded by (NS) in the enumerator	2 532
excluded by (RIG) in the enumerator	1 619 004
remaining	1 522
irreducible among them	23
exact tests in $\mathbb{Z}[x_1]$ (§6.1): NS / BC / RG	5 / 3 / 15
survivors	0

Proof of Theorem 1.2. A suitable normalisation of the shortest vector x_1 of K has its minimal polynomial among the enumerated leaves. Every irreducible leaf violates a necessary condition for being the minimal polynomial of a shortest vector of a field with a universal $\mathbb{Z}$ -form. $\square$

No number field and no maximal order is computed in degree 7. The in-enumerator tests (NS), (RIG) are the only new ingredients. Each exclusion they make on an irreducible polynomial was re-checked in exact arithmetic (**verify_strong.gp**): for (NS), the exact minimum of Var on $\mathbb{Z}[x_1] \setminus \mathbb{Z}$ by Fincke–Pohst; for (RIG), **sqshape**, i.e. factorisation of $4^7 m_\alpha((t^2 + \kappa)/4)$ over $\mathbb{Q}$. In degree 7 this re-check covers 857 103 irreducible polynomials (the other 764 433 excluded polynomials are reducible), with no disagreement. In degree 6 it covers all 15 714 exclusions on irreducible polynomials made by the strong mode, also with no disagreement. **run_degree7.sh** reproduces the computation in about 30 minutes on 4 cores.

8. CONSISTENCY CHECKS AND STATUS

- **Enumerator.** **enum6.c** agrees exactly with an exact-arithmetic re-implementation (**enum_check.gp**, real-rootedness by Sturm sequences) on all 56 subtrees and all 2 088 667 leaves in degree 6, and with brute force over coefficient boxes in degrees 3–5. In degree 7 it agrees with **enum_check.gp** on the 48 subtrees with the fewest leaves (2 267 434 leaves in total).
- **Degrees ≤ 5 reproduced.** The same pipeline, with the degree- d bound of Corollary 4.2, leaves $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{5})$ in degree 2, and $\mathbb{Q}(\zeta_9)^+$, $\mathbb{Q}(\zeta_7)^+$ in degree 3. Of these, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\zeta_9)^+$ fail (N) at $2 - \sqrt{2}$ and $2 - \zeta_9 - \zeta_9^{-1}$. Nothing is left in degree 5. In degree 4, Case A leaves nothing, and Case B leaves three fields ($d_K = 725, 1600, 7168$), all failing (N). This recovers [KY2, Thm. 1.2].
- **Controls.** (N) holds for all tested elements of $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\zeta_7)^+$; both fields survive every filter.
- **Floating point.** Used for embeddings (60 digits), for root brackets in the enumerator (80-bit, with margins), and for δ_F (double precision with a grid margin of 0.002 resp. 0.02). The grid values $\delta_{\mathbb{Q}(\sqrt{5})} \approx 1.3090 = \varphi^2/2$ and $\delta_{\mathbb{Q}(\sqrt{2})} \approx 1.7071 = 1 + 1/\sqrt{2}$ are consistent with exact values.
- **Recommended before publication.** An independent re-implementation of Case A and of the (N) searches (for example in Magma), and an exact-arithmetic computation of δ_F for the four fields.

Remark 8.1 (degree 8 and beyond). Lemma 3.2 gives $s/\gamma_s^2 \geq \frac{3}{2}$ also for $s = 7, 8$ ($\gamma_7^7 = 64$, $\gamma_8 = 2$). So budget–cost, rigidity and Corollary 4.2 hold for $d \leq 9$, and $d = 8$ has $\nu(K) \leq 3 + 2\sqrt{2} = 5.83$. Two things are new in degree 8. Case B must handle quadratic subfields with relative degree 4 and quartic subfields with relative degree 2. And the Case A enumeration is much larger: the leaf count grows by roughly 2.5–3 per unit of a_2 , and the range of a_2 grows with the bound on ν . The in-enumerator tests of §7 are what keeps the output manageable.

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